

Detecting and Quantifying Vortex-Centre Clustering in Turbulence via Spatial Statistics

Abstract

This project validates a spatial-statistics pipeline to detect clustering and non-uniformity in 2D point patterns, specifically vortex centres extracted from a turbulent velocity field. Four independent estimators are used: Voronoi tessellation, square-grid binning, K-means clustering, and nearest-neighbour distance. Each method maps spatial arrangements to characteristic distributions (cell areas, cell occupancies, point-to-centroid distances, and nearest-neighbour distances).

Empirical results are compared against analytical or Monte-Carlo complete spatial randomness (CSR) null models. Agreement is quantified using R^2 , Kolmogorov–Smirnov test statistics (D and p), coefficient-of-variation ratios, z-scores, and the Clark–Evans index (R_{CE}). The pipeline is validated on synthetic datasets before application to a 3D forced isotropic turbulence slice. All four methods consistently reject the uniform hypothesis, demonstrating significant clustering in the turbulent vortex centres.

1 Introduction

Spatial distributions of point sets contain key signatures of underlying physical mechanisms, seen in fluid particles, material defects, and turbulent structures. This project develops an integrated spatial-statistics framework to determine if a given 2D point pattern is spatially uniform or clustered, and to compute a reliable degree of confidence for this qualification.

The pipeline evaluates four core methods:

1. Voronoi tessellation
2. Square-grid binning
3. K-means clustering
4. Nearest-neighbour distance

Using a progressive validation strategy, the methods are first calibrated against synthetic controlled datasets (uniform and clustered ground truths) before evaluating real vortex centres extracted from an isotropic turbulence field.

2 Project Objective

The primary objective is to implement a robust statistical pipeline to:

1. Generate synthetic uniform and clustered point patterns.
2. Apply and compare multiple spatial-statistics estimators.
3. Evaluate empirical data against matched CSR null models using R^2 and KS tests.
4. Extract vortex structures from physical turbulence and report a standardized, numerical degree of clustering over basic visual inspection.

3 Theoretical Background

3.1 Complete Spatial Randomness (CSR)

CSR acts as the unstructured baseline. Synthetic fields are sampled uniformly over a unit square $(x_i, y_i) \in [0, 1]^2$. For physical datasets, matched CSR configurations mimic the experimental boundary domain $[0, 2\pi]^2$ and exact point counts.

3.2 Voronoi Tessellation

Voronoi cell areas offer a localized measurement of point densities. Normalized cell areas are defined by:

$$y = \frac{A}{\langle A \rangle}.$$

For uniform random Poisson fields, y follows a gamma distribution:

$$f(y) = \frac{b^b}{\Gamma(b)} y^{b-1} e^{-by}, \quad y \geq 0, \quad b = 3.57.$$

The standard reference coefficient of variation is $CV \approx 0.53$. Clustered data yields wider, right-skewed area distributions. Boundary cells are excluded from calculations to maintain finite areas.

3.3 Grid-Binning

The domain is partitioned into M equal square cells. Under a uniform random distribution, the cell point count k fits a Poisson distribution:

$$k \sim \text{Poisson}(\lambda), \quad \lambda = \frac{N}{M}.$$

The variance profile is tracked via the Fano factor:

$$F = \frac{\text{Var}(k)}{\langle k \rangle}.$$

Ideal CSR states yield $F \approx 1$, whereas clustered arrays create spatial voids and high-density cells, leading to $F > 1$.

3.4 K-Means Clustering

K-means segments the point pattern into k localized clusters. The Euclidean distance from each point to its assigned centroid serves as the structural diagnostic. Distances are normalized by their global mean. Because no closed analytical null expression exists for this metric, it relies on a simulated Monte-Carlo uniform baseline.

3.5 Nearest-Neighbour Distance

This metric evaluates the distance between every point and its nearest companion. For a 2D Poisson process with spatial intensity $\rho = N/\text{Area}$, the theoretical distance conforms to a Rayleigh-type distribution:

$$f(r) = 2\pi\rho r e^{-\pi\rho r^2}.$$

The expected mean reference distance is:

$$\langle r \rangle_{\text{CSR}} = \frac{1}{2\sqrt{\rho}}.$$

3.6 Statistical Diagnostic Notations

To assess the empirical distributions against their respective null models without explicit structural derivations, the following standardized indicators are utilized:

- **Coefficient of Determination (R^2):** Quantifies spatial agreement between empirical distributions and analytical curves or Monte-Carlo reference limits.
- **Kolmogorov–Smirnov Test (D, p):** A non-parametric distance parameter (D) and its associated probability (p) evaluating discrepancies between empirical cumulative distributions and reference states.

- **Clark–Evans Index (R_{CE}):** A dimensionless nearest-neighbour distance diagnostic where $R_{CE} = 1$ indicates randomness, $R_{CE} < 1$ represents clustering, and $R_{CE} > 1$ denotes regular spacing.
- **CSR-Based z-Score (z):** Measures the standard deviations an empirical metric (x_{real} , such as CV or mean distance) sits away from the simulated CSR ensemble mean ($\mu_{\text{CSR}}, \sigma_{\text{CSR}}$).

4 Shared Methodology

4.1 Synthetic Point Generation

Uniform generators pick points independently across the square space. Clustered generators blend uneven Gaussian clusters with a background noise fraction. Base optimization presets use $n_{\text{clusters}} = 5$, background fraction = 0.45, and a cluster spatial spread of 0.06.

4.2 Geometric and Distance Formulations

Voronoi partitions are computed over valid internal nodes, utilizing the shoelace formula for finite cell areas. Grid-binning isolates densities via 2D histograms. K-means coordinates use initializations via `scipy.cluster.vq.kmeans2` with `minit="++"`. Nearest-neighbour searches leverage `scipy.spatial.cKDTree`, using a toroidal periodic distance wrapper for physical turbulence fields.

5 Step-by-Step Methodology and Results

5.1 Voronoi Analysis of Synthetic Data

Using twin datasets of $N = 1200$ points within a unit square, Voronoi maps were generated for uniform and clustered profiles.

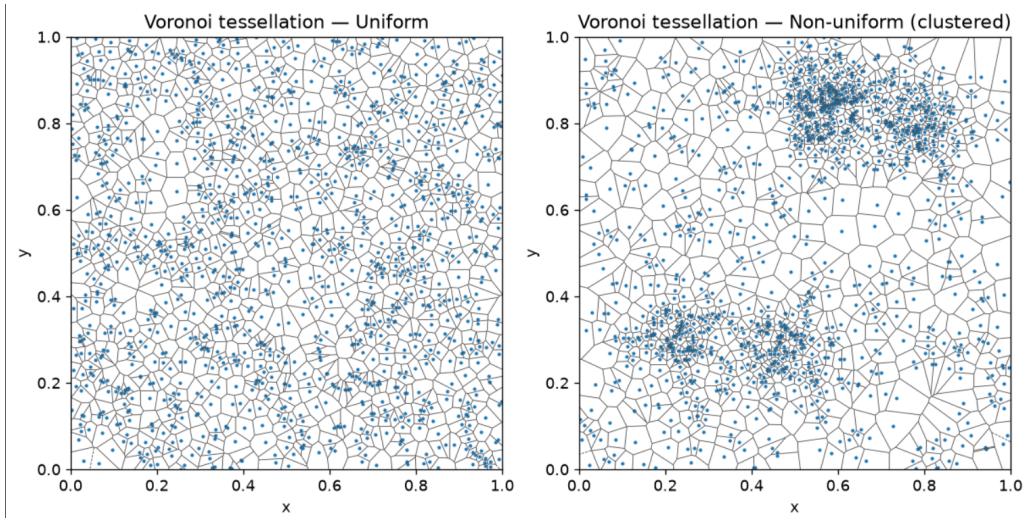


Figure 1: Voronoi tessellations for uniform and clustered synthetic point patterns.

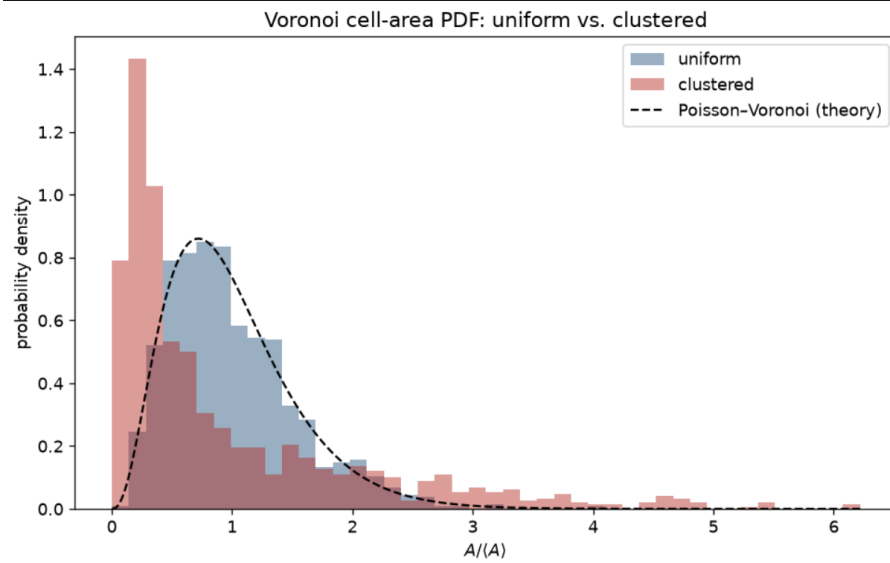


Figure 2: Probability density functions of normalized Voronoi cell area for uniform and clustered data.

Pattern	Cells	Std($A/\langle A \rangle$)	IQR	Skewness
Uniform	948	0.537	0.689	1.082
Clustered	1053	1.181	1.158	2.221

Table 1: Spread statistics of normalized Voronoi cell areas.

5.2 Grid-Binning, Nearest-Neighbour, and K-Means

Alternative diagnostics were tested against a standardized $N = 1000$ point dataset featuring four distinct Gaussian clusters and 120 background noise points. Grid-binning was applied using a 20×20 array. Nearest-neighbour distances and K-means profiles ($k = 4$) were tracked sequentially.

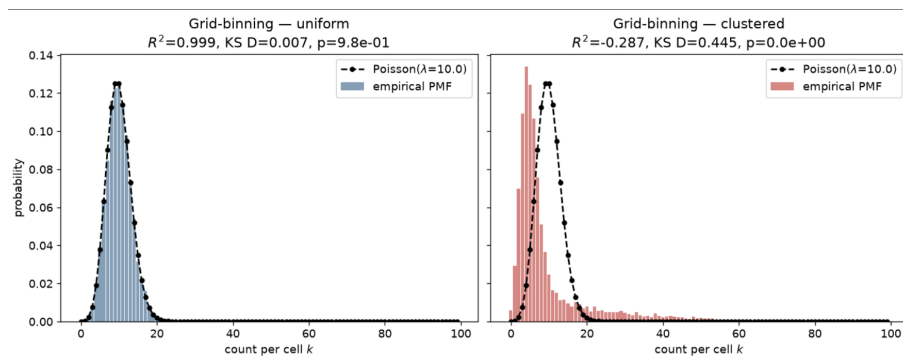


Figure 3: Normalized grid-occupancy distributions for uniform and clustered point patterns.

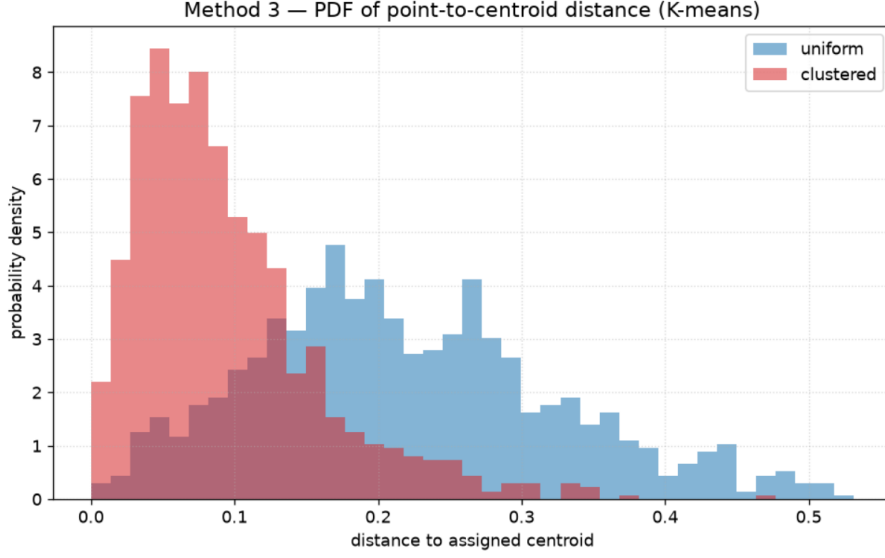


Figure 4: Point-to-centroid distance distributions for uniform and clustered synthetic data.

Diagnostic	Uniform	Clustered
Grid Fano factor	0.946	8.902
Nearest-neighbour distance CV	0.518	1.111
Mean K-means point-to-centroid distance	0.219	0.094

Table 2: Alternative spatial-estimator results.

5.3 Poisson–Voronoi Goodness-of-Fit

Data was pooled across 15 iterations ($N = 3000$ points each) to map empirical areas to the theoretical gamma model.

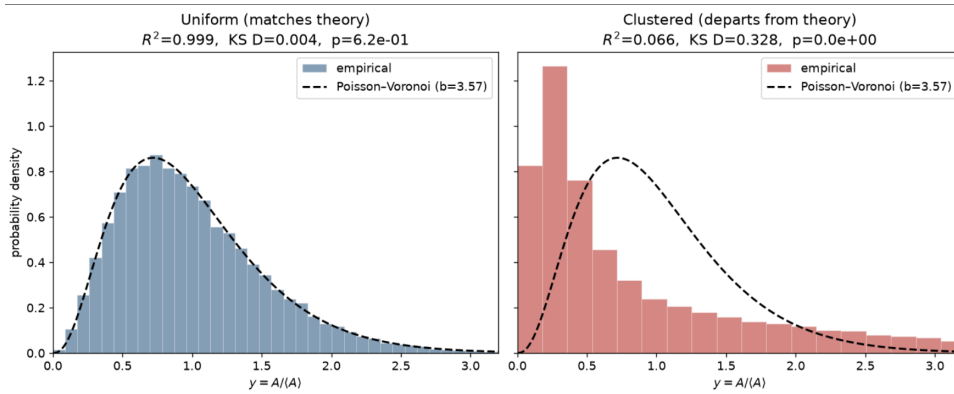


Figure 5: Goodness-of-fit comparison between synthetic Voronoi distributions and the Poisson–Voronoi gamma law.

Pattern	Cells	R^2	KS D	KS p-value
Uniform	36 220	0.9987	0.0040	0.617
Clustered	40 275	0.0662	0.3280	≈ 0

Table 3: Voronoi goodness-of-fit results.

5.4 Grid-Binning and K-Means Goodness-of-Fit

Grid counts were pooled over 20 iterations ($N = 4000$ points, 20×20 lattice, $\lambda = 10$). K-means profiles ($k = 4$) were evaluated alongside synthetic records.

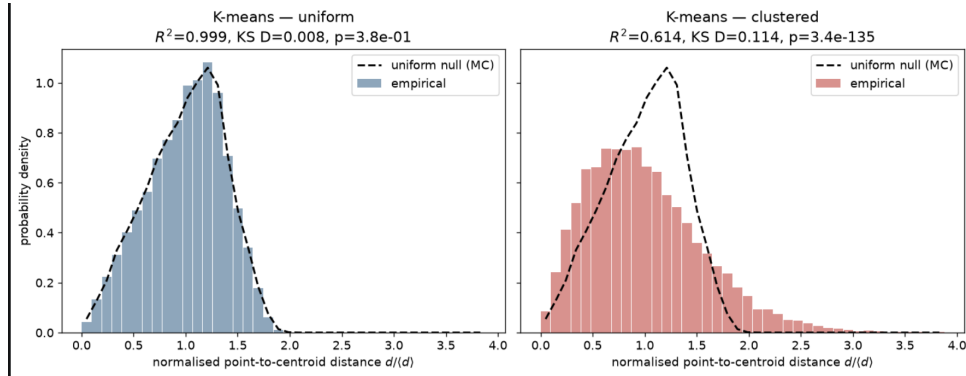


Figure 6: K-means distance goodness-of-fit for uniform and clustered synthetic patterns.

5.5 Real Turbulence Data Analysis

The pipeline evaluated out-of-plane vorticity ($\omega_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$) from an isotropic 1024×1024 turbulence slice domain ($[0, 2\pi]^2$). Vortex cores were filtered using local maxima within a 7×7 patch exceeding a high-intensity threshold:

$$T = \langle |\omega_z| \rangle + 2 \text{std}(|\omega_z|) = 19.01.$$

This isolated $N = 3186$ physical centers, compared against a 150-core simulated CSR ensemble.

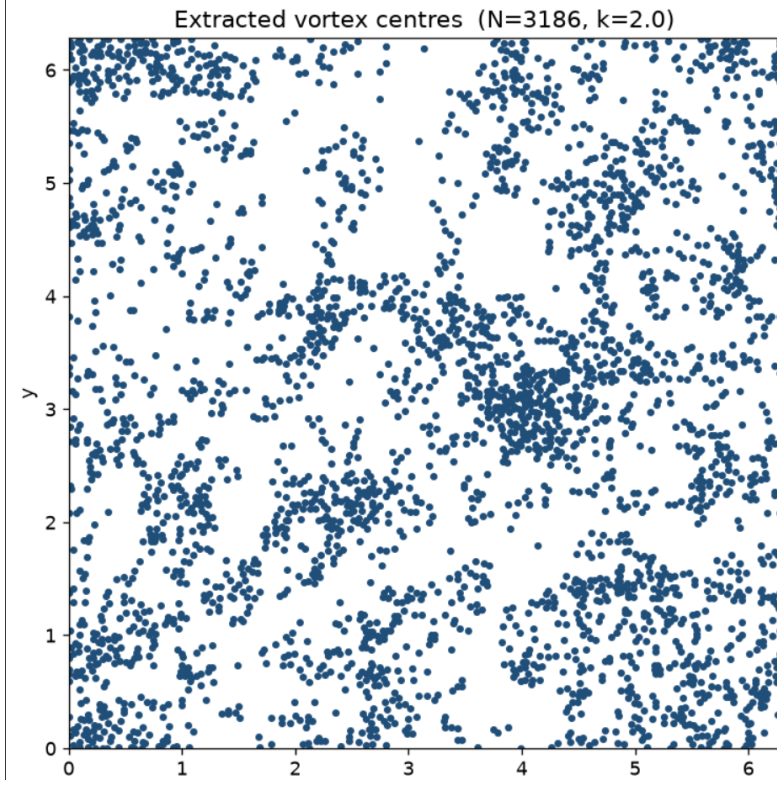


Figure 7: Detected vortex centres in the real isotropic turbulence slice.

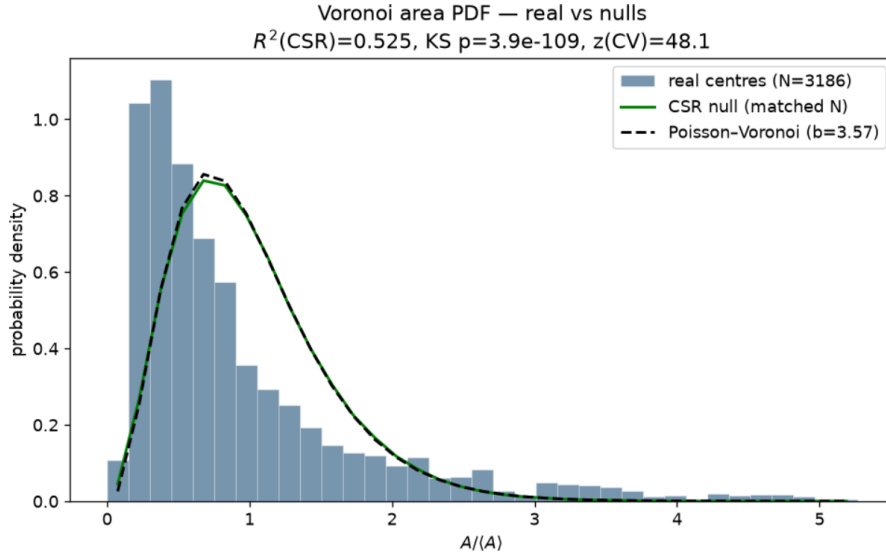


Figure 8: Voronoi area distribution of real vortex centres compared with the CSR null and Poisson–Voronoi gamma model.

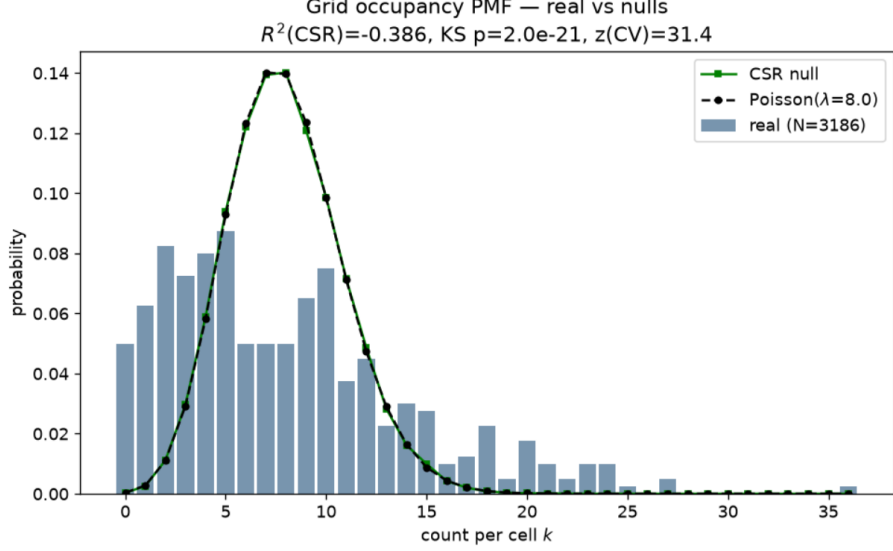


Figure 9: Grid-binning distribution of real vortex centres compared with CSR and Poisson references.

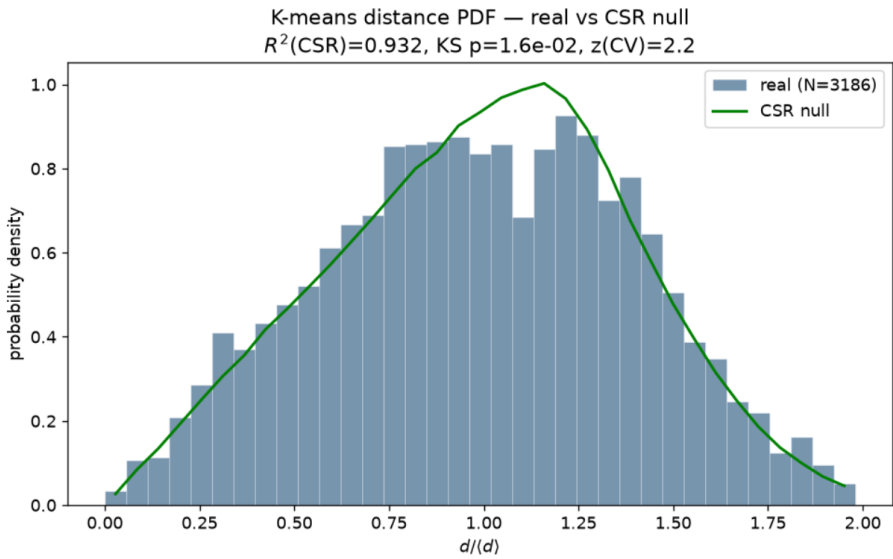


Figure 10: K-means point-to-centroid distance distribution for real vortex centres compared with the CSR null.

5.6 Nearest-Neighbour Quantified Metrics

Nearest-neighbour checking was applied to periodic domains using a 2π toroidal metric configuration. Synthetic calibration parameters are compiled below:

Case	$R^2(\text{CSR})$	KS p-value	Clark–Evans Index (R_{CE})
Synthetic uniform	0.9947	0.353	0.997
Synthetic clustered	0.608	≈ 0	0.898

Table 4: Nearest-neighbour synthetic validation results.

Evaluating the real turbulence field yielded $R_{\text{CSR}}^2 = 0.607$, $R_{\text{Rayleigh}}^2 = 0.604$, and a p_{KS} value of 7.8×10^{-70} . The tracking index yielded $R_{\text{CE}} = 0.933$, with an corresponding mean distance z-score of $z_{\text{mean}} = -7.8$. This confirms localized grouping configurations.

6 Synthetic Goodness-of-Fit Summary

The overarching performance metrics across all four controlled verification validation trials are summarized below:

Method	Uniform R^2	Uniform KS p	Clustered R^2	Clustered KS p
Voronoi area	0.9987	0.617	0.0662	≈ 0
Grid-binning	0.9989	0.978	-0.287	≈ 0
K-means distance	0.9989	0.381	0.614	3.4×10^{-135}
Nearest-neighbour	0.9947	0.353	0.608	≈ 0

Table 5: Goodness-of-fit results for all four methods on synthetic data.

7 Final Real-Data Results

The composite analytical results derived from checking the physical turbulent vortex center configurations against CSR distributions are structured below:

Method	$R^2(\text{CSR})$	$R^2(\text{Analytical})$	KS p-value	Spread Measure	z-score	Verdict
Voronoi area	0.525	0.516	3.9×10^{-109}	CV ratio = 2.00	48.1	Clustered
Grid-binning	-0.386	-0.405	2.0×10^{-21}	CV ratio = 2.14	31.4	Clustered
K-means distance	0.932	—	1.6×10^{-2}	CV ratio = 1.06	2.2	Clustered, weak
Nearest-neighbour	0.607	0.604	7.8×10^{-70}	$R_{\text{CE}} = 0.933$	-7.8	Clustered

Table 6: Final degree-of-uniformity results for the real vortex-centre point pattern.

8 Conclusion

This project successfully developed and implemented a spatial-statistics pipeline to verify point-pattern distributions. The pipeline was rigorously calibrated against synthetic benchmarks and subsequently deployed on real isotropic turbulence velocity slices to classify $N = 3186$ vortex structures. All four analytical methods decisively rejected the spatial randomness hypothesis. Voronoi and grid-binning measurements provided the strongest statistical discrepancies ($z_{\text{CV}} = 48.1$ and 31.4 respectively), while periodic nearest-neighbour analysis corroborated local pairing dynamics ($R_{\text{CE}} = 0.933$, $z_{\text{mean}} = -7.8$). The collective metrics confirm that vortex centres in forced isotropic turbulence fields are significantly clustered.